Panel C provides an estimate of the total amount of access fees collected and rebates distributed.[1114] In 2023 there were an estimated $3.41 billion in access fees collected across all exchanges and $3.08 billion in rebates distributed, resulting in a net capture to all exchanges of $340 million.

Panel D of Table 6 provides estimates of the net access fee paid by liquidity demanders and liquidity suppliers.[1115] In 2023 liquidity demanders paid an estimated $2.97 billion in net access fees and liquidity providers received an estimated $2.63 billion in rebates. The difference of $340 million is the exchanges' estimated net capture.

Although not subject to Rule 610(c), because they do not post protected quotes, ATSs also often assess transaction fees.[1116] As of the third quarter of 2023 there were 32 ATSs that

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1114 These estimates are computed by assuming a 30 mil access fee and 28 mil rebate on all transactions that occur on maker-taker or inverted exchanges and a 10 mil access fee (and 4 mil rebate) on the volume priced equal to, or greater than, $1.00 per share that occurs on IEX. For trading in sub $1.00 transactions, the various access fees and rebates for each exchange presented in Panel B of Table 4 are multiplied by the corresponding dollar volume of trade in transactions priced less than $1.00 per share to compute the total access fees collected and rebates distributed for this volume. The figures are summed together to provide the estimates of total access fees collected and rebates distributed.

1115 This estimate presumes that for shares transacted in prices equal to or greater than $1.00 per share on maker-taker venues the liquidity demander pays a 30 mil access fee and the liquidity provider receives a 28 mil rebate. On inverted exchanges the opposite occurs. On IEX it is presumed that liquidity demanders pay an 8 mil access fee and liquidity providers receive no rebate. For trading in sub $1.00 transactions the various access fees and rebates for liquidity suppliers and demanders are computed by taking the respective fees and rebates for sub $1.00 transactions for each exchange presented in Panel B of Table 4 and multiplying them by the corresponding dollar volume of trade in transactions priced less than $1.00 to compute the total access fees collected and rebates distributed for liquidity-providing and demanding trades. The figures are summed together to provide the estimates of total access fees collected and rebates distributed.

1116 IntelligentCross ATS, for example, offers matching processes for all NMS stocks eligible for trading, and disseminates bids and offers in real-time to subscribers to the ATS's proprietary data feed, but these are not protected quotes. See IntelligentCross, Form ATS-N, Item 15 (Display) (dated Apr. 11, 2022) available at https://www.sec.gov/Archives/edgar/data/1708826/000170882622000002/xslATS-N_X01/primary_doc.xml.

reported trading volume to FINRA transacting a total of 73 billion shares.[1117] Unlike exchanges, the fees that ATSs charge generally do not have a standard structure and are often negotiated between the ATS and the customer. Based on a review of item 19 in form ATS-N, ATSs generally do not provide rebates, and when transaction fees are explicitly discussed, they are often in the range of 10 mils.[1118]

Table 4 indicates that many exchanges charge the maximum allowed fee, rebating nearly all of it as a compensation for liquidity provision. One commenter states, "access fees have been uniquely impervious to market forces."[1119] Rule 611 generally causes marketable orders to be routed to those markets displaying the best-priced quotations.[1120] As discussed in section VII.B.3, a liquidity provider is generally indifferent between receiving compensation in the form of a rebate or in the form of a quoted spread, implying that an exchange can use rebates to induce quoting lower spreads, and hence the best prices.[1121] The exchange can fund the rebate with an

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1117 See FINRA, ATS Transparency Data Quarterly Statistics, available at https://www.finra.org/filing-reporting/otc-transparency/ats-quarterly-statistics.

1118 See infra note 1442 for commenter discussion on ATS transaction fees. See also IEX Letter VI at 5 for additional analysis supporting the conclusion that 10 mils is a representative transaction fee among ATSs.

1119 See IEX letter V at 2, 3. See also Proposing Release, supra note 11, at 80305.

1120 More specifically, Rule 611 requires trading centers to have policies and procedures that reasonably prevent trade-throughs. See Proposing Release, supra note 11, at 80286. A trade-through is a trade that executes at a price lower than a protected bid or higher than a protected offer. The NBBO is set by the best protected bid and offer; therefore, a trade that executes outside the NBBO is a "trade-through." If an exchange does not have a limit order at the best quote, then it cannot execute against an incoming marketable order; rather the exchange would generally need to cancel the order or route it to another exchange with the best quote. The routing of marketable orders is prevalent. A recently published academic article finds that 34% of market orders sent to the NYSE in 2010-11 are routed. See Sida Li et.al., Refusing the Best Price? 2 J. FIN. ECON. 147 (February 2023). For example, suppose two liquidity providers want to sell a share in exchange for $10.002, net of fees and rebates. Suppose the first seller posts at exchange X, which offers a 30mil rebate, while the second seller posts at exchange Y, which offers a 10mil rebate. The seller on exchange X is willing to quote at $10.00 to receive a net price of $10.003, while the seller on Y is not willing to quote at $10.00 because the net price would be only $10.001—the seller on Y must quote at $10.01. Rule 611 will therefore direct marketable orders to the lower quoted price at exchange X.

1121 See Proposing Release, supra note 11, at 80305 ("the NBBO restricts the routing behavior of marketable orders and often forces liquidity demanders to pay the access fee to trade against a NBBO order. Exchanges access fee charged to the liquidity demander, relying on Rule 611 to reduce the loss of liquidity demanding customers that would otherwise occur from such an increase in prices.[1122] The exchanges profit from the difference between the access fees collected and the rebates paid. Were exchanges to unilaterally lower their access fees and rebates (without other exchanges making similar changes), liquidity providers would likely route their orders to another exchange.[1123] Notably, research surrounding a Nasdaq experiment where it unilaterally lowered fees and rebates found that Nasdaq lost market share to other maker-taker venues with a higher rebate.[1124] Table 4 also shows that even the maximum rebates are close to the access fees; doing otherwise would likely be unprofitable or risky.[1125]

As discussed in the Proposing Release, and as Table 4 shows, the NYSE, Nasdaq, and Cboe exchange families each operate both a maker-taker venue as well as an inverted venue,[1126] Commenters state that inverted venues can be used to achieve intra-tick pricing.[1127] Specifically,

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are thus incentivized to attract more competitively priced liquidity with large rebates, which are funded by similarly large access fees, in order to capture more trading volume"). The high rebate allows the liquidity supplier to offer a better quoted price—i.e., a higher bid or a lower offer—because the liquidity supplier only cares about the total proceeds from the sale (the liquidity supplier does not care whether the proceeds take the form of a rebate).

1122 That is, the need to execute against the protected quote first, before executing at other prices, would maintain a strong incentive for broker-dealers to route orders to the exchange, even in the face of high access fees.

1123 See Proposing Release, supra note 11, at 80305 n.457.

1124 See id.; see also Yiping Lin, et al., A Model of Maker-Taker Fees and Quasi-Natural Experimental Evidence (working paper Feb. 8, 2021), available at https://ssrn.com/abstract=3279712 (retrieved from SSRN Elsevier database). Consequently, it could be harmful to an exchange to unilaterally reduce access fees and their associated rebates if other exchanges do not follow suit. Further, even if each of the exchanges lowered its fees, there would be the risk that a new exchange would see the opportunity and enter the market with high fees and rebates and thus capture market share, inducing the other exchanges to abandon their low fee models to remain competitive.

1125 See discussion in section VII.D.2.b.

1126 See Proposing Release, supra note 11, at 80313.

1127 See Larry Harris, Quarter Penny Tick (working paper, Mar. 9, 2022) attached to Letter from Larry Harris ("Quarter Penny Tick").

the net price of a trade is the quoted price adjusted for exchange fees and rebates; the quoted price is constrained by the tick, but the net price can be between ticks.[1128] To the extent that intra-tick pricing on inverted venues is a solution to the quoted price being constrained by the tick, it is a costly one. First, quote protection applies to the quoted bid or ask, not the net cost, implying that routing a market order to an inverted venue runs the risk of the order being routed elsewhere if the inverted venue is not at the NBBO. Research shows that inverted venues are less likely to be at the NBBO, a result that follows from revenue from rebates and from spreads being interchangeable assuming the market participant receives both.[1129] As explained in section VII.D.1.b.ii, this leads to delays and increased cost.

Second, the existence of inverted venues fragments liquidity compared to the situation where an exchange is able to offer orders to be placed at multiple prices within the quoted spread. One commenter agreed when discussing inverted venues, stating that: "different exchanges are optimal to use for different prices within the penny, which is a recipe for artificial fragmentation, a confusing paper trail, and overall excess complexity. Excess complexity, in turn, is a recipe for excess rents, agency conflict and distrust."[1130]

Lastly, one commenter stated that exchanges could circumvent barriers associated with the tick size and access fees through innovation, such as new order types.[1131] The commenter

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1128 Suppose an exchange family operates both a maker-taker venue and an inverted venue; further suppose that the maker-taker venue offers a 30mil rebate to liquidity suppliers, while the inverted venue charges a 30mil fee to liquidity suppliers. Liquidity suppliers would therefore be able to transact at two different net prices within the tick—e.g., the liquidity supplier could offer to sell at $10.00 on the maker-taker venue, which would result in a net price of $10.003; or the liquidity supplier could offer to sell at $10.00 on the inverted venue to net $9.997. The variation in fee schedules across the venues therefore allows for intra-tick pricing. See also supra section VII.C.2.b for a discussion of current state of the fees and rebates and the variation in pricing structure across exchanges.

1129 See IEX Letter I at 14.

1130 See Budish Letter at 4.

1131 See Virtu Letter II at 23.

stated that: "Lastly, if the tick size were really a significant barrier to competition for exchanges, they could innovate solutions to solve for this. For example, exchanges could develop an order type that functions within the current structure (limit order pricing and priority-ranked based on even penny ticks) but where an order could provide sub-penny price improvement if matched to a marketable order from a counterparty that met certain objective conditions (such as being sourced from a retail customer)."[1132] The commenter proceeded to describe a second novel order type that could allow for sub-penny price improvement through reduced access fees for retail investors.[1133] The order types the commenter lists as examples appear to solve for the specific problem of retail investors achieving price improvement on exchange, a solution that already exists through RLPs (though which do not have significant volume).[1134] It is possible that a new order type designed to mitigate this problem might not work as intended. In contrast, allowing for more ticks in certain stocks as the Commission is adopting is a more straightforward and predictable way of achieving the same ends without requiring the need for order types designed to allow for sub-tick executions on exchanges.

3. Round Lots, Odd-Lots, and Market Data Infrastructure

Currently, information on odd-lot quotes inside the NBBO is available only to investors who subscribe to proprietary data feeds, and comprehensive odd-lot information is only available to market participants who subscribe to the proprietary data feeds of all the exchanges. The implementation of the MDI Rules will include odd-lot information inside the NBBO.[1135] The

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1132 See id.

1133 See id.

1134 See supra section VII.C.1.a for further discussion of exchange RLP programs.

1135 See supra section VI.C and section V.E for discussions on the expected time of the implementation of the MDI Rules.

MDI Rules also defined a round lot, which previously had not been defined in a Commission rule. Specifically, the MDI Rules establish a uniform round lot size of 100 shares for stocks priced $250 or less; 40 shares for stocks priced greater than $250 and less than or equal to $1,000; 10 shares for stocks priced greater than $1,000 and less than or equal to $10,000; finally, 1 share for stocks priced greater than $10,000. These amendments modify the round lot definitions set by the MDI Rules by changing the evaluation period in which a stock's share price is measured. The MDI Adopting Release defined round lots based on the stock's average price in the preceding month—i.e., a stock's round lot was updated every month based on the most recent month's data. These amendments update round lots every six months, with the round lot determined by the stock's average price with a one-month lag—i.e., a stock's round lot is updated in May of every year using its average stock price in March, and the round lot is updated again in November using its average stock price in September.

In the MDI Adopting Release, the Commission established a transition period for the implementation of the MDI Rules.[1136] The Commission's approval of the MDI Plan Amendments will be the starting point for the rest of the MDI implementation schedule.[1137] After approval of the MDI Plan Amendments, the next step will be a 180-day development period, during which competing consolidators can register with the Commission.[1138] Based on the times provided in the transition plan for implementation of the MDI Rules, the Commission estimated

1136 See MDI Adopting Release, supra note 10, at 18698-18701.

1137 See id. at 18698.

1138 See id. at 18699-18700.

that the full implementation of the MDI Rules will be at least two years after the Commission's approval of the plan amendment(s) required by Rule 614(e).[1139]

The Operating Committees of the CTA/CQ Plan and UTP Plan filed the MDI Plan Amendments on November 5, 2021.[1140] The Commission disapproved the proposed amendments on September 21, 2022.[1141] As a result, the participants to the effective national market system plan(s) will need to develop and file new proposed amendments as required by Rule 614(e),[1142] before the implementation period prescribed by the phased transition plan can commence. Because the implementation of the MDI Rules has been delayed, the end date of the implementation period cannot be estimated with certainty.

The following discussion reflects the Commission's assessment of the anticipated economic effects of the MDI Rules described in the MDI Adopting Release as they relate to the baseline for the adoption of these amendments.[1143] The MDI Rules are part of the regulatory baseline for this rule because they have been adopted. Given that the MDI Rules have not yet been implemented, they have not affected market practice and therefore data that would be required for a quantitative analysis of a baseline that includes the effects of the MDI Rules is not

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1139 See id. at 18700-18701.

1140 The Operating Committees of CTA Plan and UTP Plan filed proposed amendments on Nov. 5, 2021, which were published for comment in the Federal Register. See Securities Exchange Act Release Nos. 93615 (Nov. 19, 2021), 86 FR 67800 (Nov. 29, 2021); 93625 (Nov. 19, 2021), 86 FR 67517 (Nov. 26, 2021); 93620 (Nov. 19, 2021), 86 FR 67541 (Nov. 26, 2021); 93618 (Nov. 19, 2021), 86 FR 67562 (Nov. 26, 2021).

1141 See Securities Exchange Act Release Nos. 95848 (Sept. 21, 2022), 87 FR 58544 (Sept. 27, 2022); 95849 (Sept. 21, 2022), 87 FR 58592 (Sept. 27, 2022); 95850 (Sept. 21, 2022), 87 FR 58560 (Sept. 27, 2022); 95851 (Sept. 21, 2022), 87 FR 58613 (Sept. 27, 2022).

1142 The Commission ordered the exchanges and FINRA to file a new plan regarding consolidated market data on Sept. 1, 2023. On Jan. 19, 2024, the Commission published notice of filing of a National Market System Plan for Consolidated Equity Market Data. See supra note 78.

1143 See MDI Adopting Release, supra note 10, at 18741-18799.

available. It is possible that the baseline for this rule, and therefore the economic effects relative to the baseline, could be different depending on how the MDI Rules are implemented.[1144]

When adopting the MDI Rules, the Commission enumerated numerous economic effects specifically related to changing the round lot definition and including odd-lot information as a part of core data. For the change in the definition of round lots, these effects included: (1) a mechanically tighter NBBO for higher priced stocks due to the redefinition of the round lot sizes,[1145] (2) increased transparency and better order execution,[1146] and (3) potentially more orders for high priced stocks being routed to exchanges instead of ATSs.[1147] The costs of changing the round lot definition included upgrading systems to account for additional message traffic, and modifying and reprogramming systems.[1148] The Commission also discussed the expected effect that changing the round lot definition will have on other rules and regulations.[1149]

For the inclusion of odd-lot information inside the NBBO in core data,[1150] these effects include reducing information asymmetries between investors who currently have access to odd-lot information through proprietary data feeds and those who do not, leading to better order

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1144 Commission staff will review and study the effects of the amendments adopted herein. See the introduction to section VII.D.

1145 See MDI Adopting Release, supra note 10, at 18743 for the full discussion of the effect of changing the round lot size on the NBBO.

1146 See MDI Adopting Release, supra note 10, at 18744, 18747 for the full discussion of the effect of changing the round lot size on transparency and execution quality.

1147 See MDI Adopting Release, supra note 10, at 18747 for the full discussion of the effect of changing the round lot size on exchange competition and order routing.

1148 See MDI Adopting Release, supra note 10, at 18748 for the full discussion of the expected costs of changing the round lot size.

1149 See MDI Adopting Release, supra note 10, at 18749 for the full discussion of the effect of changing the round lot size on other rules and regulations.

1150 See MDI Adopting Release, supra note 10, at 18753 for the full discussion of the effect of including odd-lot information inside the NBBO in its definition of core data.

execution and price efficiency.[1151] Providing an alternative to proprietary data for some market participants will allow these market participants to reduce data expenses required for trading.[1152] The costs of including odd-lot information inside the NBBO include:[1153] the cost of upgrading existing infrastructure and software to handle the dissemination of additional core data message traffic; the cost to SROs to implement system changes required in order to make regulatory data and other data needed to generate consolidated market data available to competing consolidators; the cost of technological investments market participants might have to make in order to receive the new core message traffic; and the cost to users of proprietary data whose information advantage will dissipate somewhat.[1154]

The MDI Rules do not require the competing consolidators to disseminate odd-lot information. However, the Commission estimated that at least one competing consolidator will disseminate the odd-lot information because the Commission believed that there will be demand for the data.[1155]

4. Affected Entities and Markets

The amendments will affect trading in NMS stocks, particularly either on exchanges that charge high access fees or in stocks with lower quoted spreads, many odd-lots inside the spread, or higher prices. Therefore, the amendments will affect a wide variety of market participants, including national securities exchanges, other trading venues, exclusive SIPs and their data users,

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1151 Id.

1152 Id.

1153 See MDI Adopting Release, supra note 10, at 18759 for the full discussion of the costs associated with expanding core data to include odd-lot information inside the NBBO.

1154 Id.

1155 See MDI Adopting Release, supra note 10, at 18752 n.1945 and surrounding text.

any future competing consolidators, broker-dealers operating order entry and order routing systems, and others who engage in the trading of NMS stocks, including investors.[1156]

There are 16 national securities exchanges on which NMS stocks are traded that will be affected by the amendments. The exchanges compete with each other and other trading venues to attract order flow. Exchanges compete by setting the rules that dictate how orders routed to them interact given the broader requirements of the Exchange Act and rules thereunder. Such rules are coded into the systems of exchanges that match buy and sell orders. Exchanges also compete via their services and fee structures; they differentiate themselves with the access fees they charge or the rebates they pay out for particular order types, as well as their data and connectivity options.[1157] A subset of national securities exchanges, the five listing exchanges, also compete to attract stock listings by setting rules for listing standards for securities. The listing exchanges are also responsible for tracking certain regulatory information regarding their listed stocks.

Other trading venues, including 33 ATSs and 238 other FINRA members, including OTC market makers, also compete with exchanges and each other to attract order flow in NMS stocks and can route orders to the various trading venues. The order flow they attract depends on a number of factors such as fees and price improvement over the NBBO, services such as order display features, segmentation of subscriber order flow and the ability of subscribers to select which category of order flow to interact with, among other aspects of execution quality.

1156 According to the 2022 Survey of Consumer Finances, available at https://www.federalreserve.gov/econres/scfindex.htm?mod=article_inline, out of a total number of households of approximately 131,000,000, 58% invested in equities in some fashion (e.g., held stock directly, invested in a stock mutual fund, etc.). Bd Gov. Fed. Res., Changes in U.S. Family Finances from 2019 to 2022: Evidence from the Survey of Consumer Finances (Oct. 2023) at 19, available at financeshttps://www.federalreserve.gov/publications/files/scf23.pdf.

1157 Exchanges can also facilitate the routing of orders to other exchanges.